Guardian Point-Defence Fire Control

Detailed Experimental Report: Predictive Tracking and Turret Control for a Maneuvering Aerial Target (ME 319, Assignment 2)

[Author Name]
Department of Mechanical Engineering, IIT Bombay
github.com/VatshVan/Predictive-Tracking-and-Turret-Control-for-a-Maneuvering-Aerial-Target

November 2026

Abstract

This report documents every experiment run on the Guardian fire-control design. For each test it records what was done, what was measured, what was observed, and what that means for the design thesis. The thesis has three parts. The prescribed $t_p = 0.5\,\mathrm{s}$ prediction is a lag to be *cancelled*, not a lead. A predictor and a reference generator sharing the Taylor polynomial $A_3(s)$ cancel it exactly. And a jerk-state estimator, a plant-inverse feed-forward and a calibrated firing rule then let one frozen design engage any admissible V-44. Over 2 900 independent engagements, the frozen design intercepted every target, bounding its failure probability below 0.13% (95% confidence). Of 9 197 pulses in the main campaign, 99.96% hit. Interception time follows a slew-limited physical model ($R^2 = 0.93$). The estimator is statistically consistent ($E[z^2] = 1.008$), and the hit probability that drives firing is calibrated. Every design choice is justified by paired, multiplicity-corrected tests, including the ones that went against our expectations.

Contents

1 Thesis and Hypotheses **3**

2 System Under Test **3**

3 Analytical Results **4**
3.1 Plant and feedback loop 4
3.2 Prediction-matched reference generator 5

4 Experimental Design **6**
4.1 Simulation protocol 6
4.2 Standardized and stress scenarios 7
4.3 Monte Carlo engagements 8
4.4 Statistical methodology 10

5 Results I: Individual Scenarios **10**

6 Results II: Monte Carlo Campaign (1 000 engagements) **15**
6.1 Interception and engagement accuracy 15
6.2 Pointing accuracy over time 17
6.3 Behaviour across radar loss 19
6.4 Estimator consistency and firing calibration 20

7 Results III: Design Decisions under Paired Statistics **21**
7.1 One-factor ablations (500 paired seeds) 21
7.2 Stress-scenario noise ensembles 22

8 Results IV: Controller Rate, Timing and Implementation **23**

9 Results V: Robustness to the Evaluation Interface **24**

10 Final Evaluation: Expected Performance on Unseen Cases **25**
10.1 What the hidden cases share with our tests 25
10.2 Quantitative expectations 26
10.3 Interface requirements the evaluation script must meet 26
10.4 Where the evidence stops 26

11 Verdicts and Conclusions **27**

A Reproducing This Report **27**

1 Thesis and Hypotheses

The laser is instantaneous, so a hit requires the turret to point at the target's *current* position at the instant of the pulse. The assignment nevertheless prescribes a predicted elevation reference at $t_p = 0.5\,\mathrm{s}$. Our thesis reconciles the two:

> *The t_p prediction exists to cancel the lag of the turret loop. If the predictor $A_3(\mathcal{D})$ and the reference generator share the same Taylor polynomial, the cancellation is exact by the time-shift property of LTI systems, so the turret tracks the target's present elevation. With a jerk-state estimator (matched to the plant's relative degree three), an exact plant-inverse feed-forward and a calibrated hit-probability firing rule, a single frozen design can engage any admissible V-44 trajectory, including through radar loss.*

We test six hypotheses that the thesis implies:

H1 Cancellation. With the prediction-matched reference generator the turret tracks the now-cast. Pointing error is set by the estimator, not by the t_p lead or the loop lag.

H2 Consistency. The estimator's covariance is honest, so the hit probability used for firing is calibrated.

H3 Estimator order. A jerk-state (order-4) estimator is necessary for maneuver switches and is the robust choice for unseen maneuvers.

H4 Firing. The fire-now-or-wait rule wastes almost no pulses and does not delay interception unnecessarily.

H5 Measurement loss. Through blackouts up to 0.5 s the design degrades gracefully, never fires blind late in a loss, and recovers quickly.

H6 Generalisation. The frozen design generalises to unseen admissible engagements *and* to the different conventions an unseen evaluation script may use.

Section 11 returns a verdict on each.

2 System Under Test

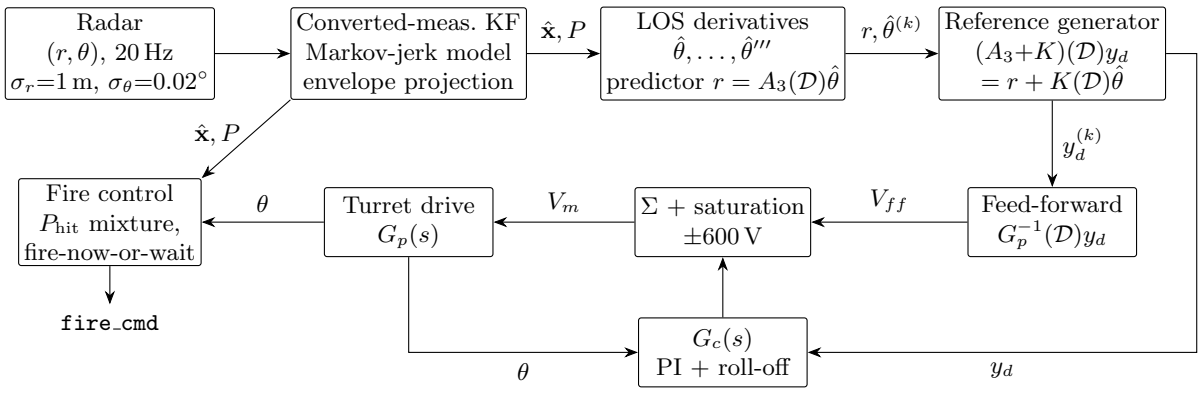


Figure 1: Architecture of the frozen design. The reference generator contains the predictor polynomial A_3, so the t_p lead and the generator lag cancel.

Table 1: Frozen design parameters (identical for every experiment unless a one-factor ablation changes one of them).

Block	Configuration
Plant	$G_p(s) = 1000/[s(320s^2 + 3350s + 1.0015 \times 10^6)]$; poles 0, $-5.23 \pm 55.70j$ (8.90 Hz, $\zeta = 0.094$); supply ± 600 V
Estimator	Unbiased converted measurements; per axis $[p, v, a, j]$, jerk Gauss–Markov $\sigma_j = 20\,\mathrm{m/s^3}$, $\tau_j = 0.4$ s; projection $\lvert\mathbf{v}\rvert \le 157.5$, $\lvert\mathbf{a}\rvert \le 23$; χ^2 jump gate 40; class re-identification after loss
Predictor	$r = A_3(\mathcal{D})\hat{\theta}$, $A_3(s) = 1 + t_p s + (t_p s)^2/2 + (t_p s)^3/6$, $t_p = 0.5$ s
Reference generator	$(A_3 + K)(\mathcal{D})y_d = r + K(\mathcal{D})\hat{\theta}$, $A_3 + K = \frac{t_p^3}{6}(s + 20)^3$; cascade gains $(p/3, p, 3p)$; rate and acceleration limits 0.58 rad/s, 2.5 rad/s^2
Feed-forward	$V_{ff} = 0.32\,\dddot{y}_d + 3.35\,\ddot{y}_d + 1001.5\,\dot{y}_d$
Feedback	$G_c(s) = 4000(1 + 1/s)/(0.03s + 1)$; conditional-integration anti-windup
Fire control	17 maneuver classes (nominal; $\lvert\mathbf{a}\rvert = 10, 20\,\mathrm{m/s^2}$ in 8 directions), nominal weight 0.5. Tracking: fire if $P_{\mathrm{hit}} \ge 0.99$, or arm at 0.90 and fire at the peak (drop 0.02) or after 0.3 s. Coasting: fire only while $P_{\mathrm{hit}} \ge 0.95$ and falling. Consensus $\ge 80\%$ of classes; ≥ 8 radar updates; 1 s recharge

3 Analytical Results

3.1 Plant and feedback loop

What we did. We derived the elevation drive from first principles, computed the open-loop Bode response of $G_c G_p$ (continuous, and with the implemented discrete compensator and a ZOH plant at 100 and 20 Hz), and simulated the plant's voltage step response and the feedback-only closed loop.

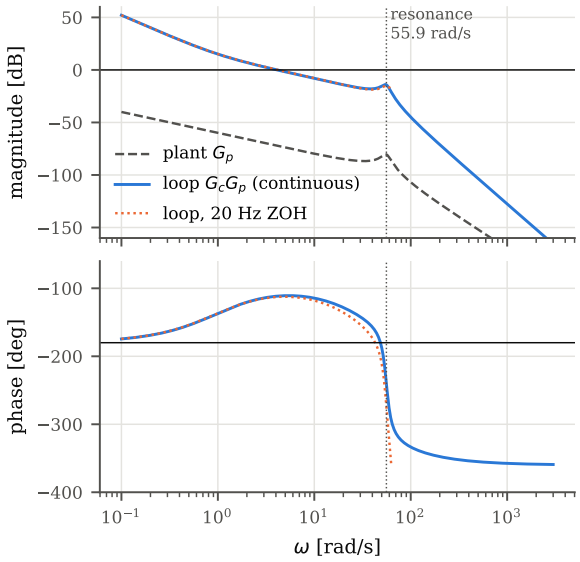


Figure 2: Bode plot of the plant G_p and the loop $G_c G_p$, continuous and with 20 Hz ZOH sampling. The dotted line marks the 55.9 rad/s drive resonance.

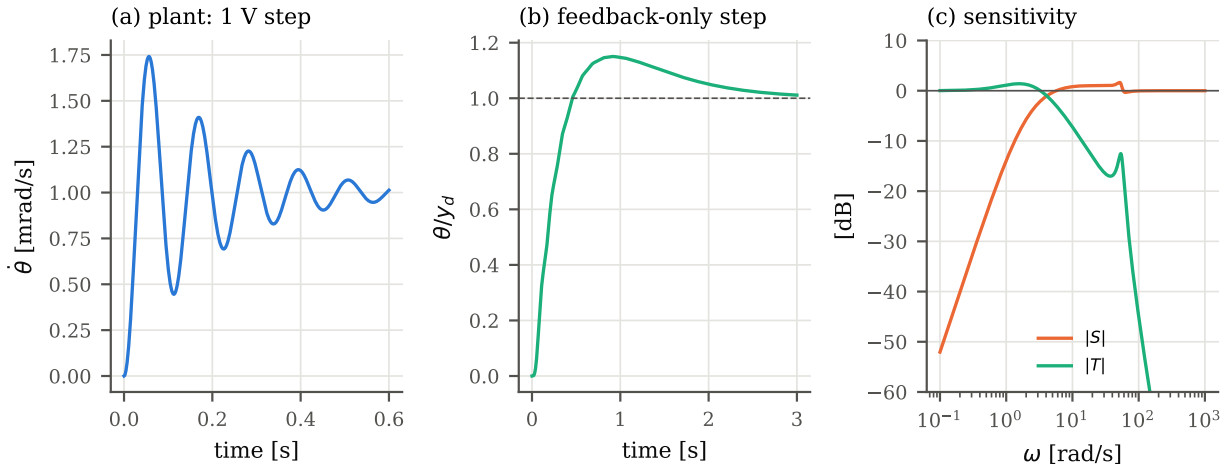


Figure 3: (a) Turret rate response to a 1 V step: the 8.9 Hz electromechanical mode rings with $\zeta = 0.094$. (b) Step response of the feedback loop alone (no feed-forward). (c) Sensitivity $|S|$ and complementary sensitivity $|T|$.

Observations.

- The plant velocity response (Fig. 3a) rings at 8.9 Hz with very light damping and settles to about 1.0 mrad/s per volt. Tracking the largest LOS rate in the envelope (150/800 = 0.19 rad/s) therefore needs about 190 V, and a ±600 V supply gives about 3× headroom.
- The loop crosses over at 4.1 rad/s with 68.5° phase margin. The resonance peak sits 16.1 dB below 0 dB (Fig. 2), so the mode is gain-stabilised without a notch.
- Sampling barely matters. Margins with the implemented discrete compensator are 68.5°/16.1 dB at 100 Hz and 67.4°/18.6 dB at 20 Hz (Table 2).
- The peak sensitivity is $M_s = 1.21$ (1.7 dB), a very robust loop. The feedback-only step response (Fig. 3b) is well damped, with no visible excitation of the 8.9 Hz mode.

What it shows. The feedback loop is deliberately modest and robust. Tracking performance comes from the feed-forward path, not from loop bandwidth (supports H1).

Table 2: Loop margins of the implemented compensator with ZOH plant.

Sampling	ω_c [rad/s]	Phase margin [deg]	Gain margin [dB]
continuous	4.10	68.5	16.1
1000 Hz	4.10	68.5	16.0
100 Hz	4.12	68.5	16.1
50 Hz	4.14	68.4	16.3
20 Hz	4.19	67.4	18.6

3.2 Prediction-matched reference generator

What we did. We computed the roots of the Taylor sections $A_n(s) = \sum_{k \le n} (t_p s)^k / k!$ of $e^{s t_p}$ for $n = 1, \ldots, 6$. We then compared the transient of the bare prediction-matched lag $1/A_3(s)$ with that of our deviation-feedback design $1/(A_3 + K)(s)$.

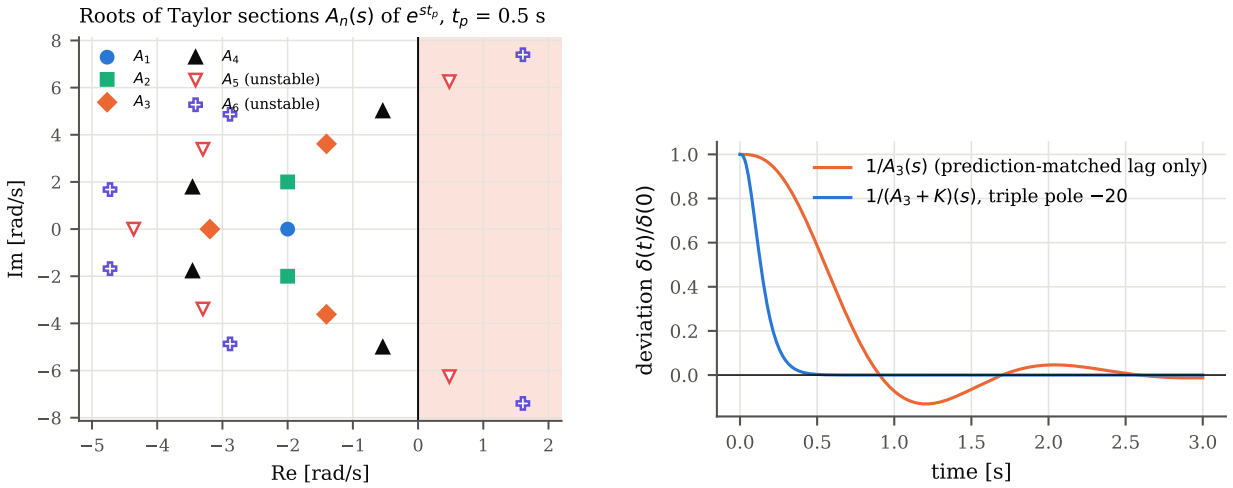


Figure 4: Left: roots of the Taylor sections $A_n(s)$. Sections 5 and 6 have right-half-plane roots. Right: decay of a unit deviation $\delta = y_d - \hat{\theta}$ with the bare prediction-matched lag $1/A_3$ and with the deviation-feedback design (triple pole at -20 rad/s).

Observations.

- A_1–A_4 are Hurwitz; A_5 and A_6 are not (Fig. 4, left). The least-damped root has $\zeta = 1.00, 0.71, 0.36, 0.11$ for $n = 1, \ldots, 4$.
- A reference model that exactly cancels a Taylor predictor therefore exists only up to fourth order, and it becomes progressively more oscillatory as the order rises.
- The bare $1/A_3$ lag decays slowly and overshoots by about 13% (Fig. 4, right). In our first prototype, which used it directly, a 0.4 rad acquisition slew overshot the target by 46 m and pointing RMS was 5.3 m.
- With deviation feedback $K(s) = 165.7 + 24.5s + 1.125s^2$, the deviation decays without overshoot in about 0.3 s, while $y_d \to \hat{\theta}$ remains exact.

What it shows. Exact cancellation of the t_p lead is possible and is bounded at order four. The deviation-feedback design keeps the cancellation while fixing its poor damping (supports H1 and motivates H3).

4 Experimental Design

4.1 Simulation protocol

The simulator mirrors the evaluation interface. At each controller step t_k:

1. it calls `target_motion_generator`(t_k);
2. it calls `turret_controller`, with the turret state;
3. it judges a pulse on every $0 \to 1$ edge of `fire_cmd` at least 1 s after the previous pulse: a hit if the target is within 2 m of the beam ray;
4. it integrates `turret_plant` with RK4 at 1 ms under a zero-order-hold voltage, with mechanical stops at $-10°$ and $+85°$.

Unless stated, the controller runs at 100 Hz and the turret starts at rest at 0°.

Metrics:

- *interception success*: at least one hit;
- *first-hit time* (the spec's interception time);

- *pulse hit rate*;
- *pointing RMS*: cross-range $r\sin(\theta - \theta_T)$ over $t \geq 2\,\mathrm{s}$;
- *time within* R_{eng};
- *peak voltage*.

4.2 Standardized and stress scenarios

What we did. Besides the specification's scenarios A–C, we built six stress scenarios as plausible hidden cases, each targeting one weakness of a predictive fire-control system (Figs. 5–6):

- **D1** *stealth break*: a $(9, 17)\,\mathrm{m/s^2}$ pull-up that starts *inside* a blackout at 3.0–3.5 s, the V-44's described tactic.
- **D2** *weave*: near-$a_{\max}$ sinusoidal jinking, the worst case for a 0.5 s predictor.
- **D3** *close pass*: the target closes to 853 m with elevation rising to 83°, giving peak LOS rate and acceleration near the elevation limit.
- **D4** *repeated loss*: scenario B with four blackouts.
- **D5** *combined*: weave plus a blackout at short range.
- **D6** *jerk test*: $\pm 18\,\mathrm{m/s^2}$ reversals every 1.2 s with 0.15 s transitions ($450\,\mathrm{m/s^3}$).

Every trajectory is generated on a 1 ms grid and checked against the envelope ($800 \leq r \leq 2000\,\mathrm{m}$, $|\mathbf{v}| \leq 150\,\mathrm{m/s}$, $|\mathbf{a}| \leq 20\,\mathrm{m/s^2}$).

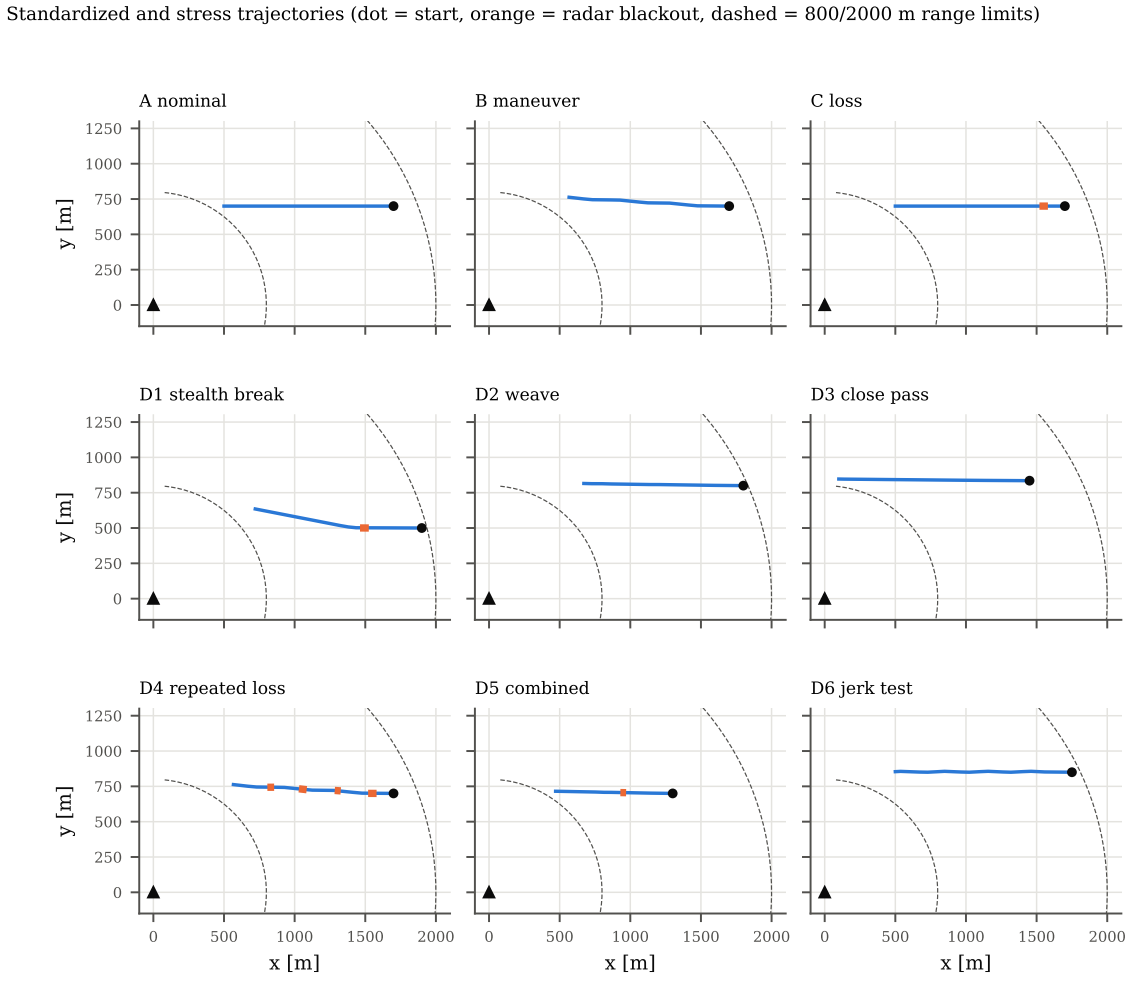


Figure 5: Trajectories of the nine scenarios in the elevation plane. Orange segments are radar blackouts.

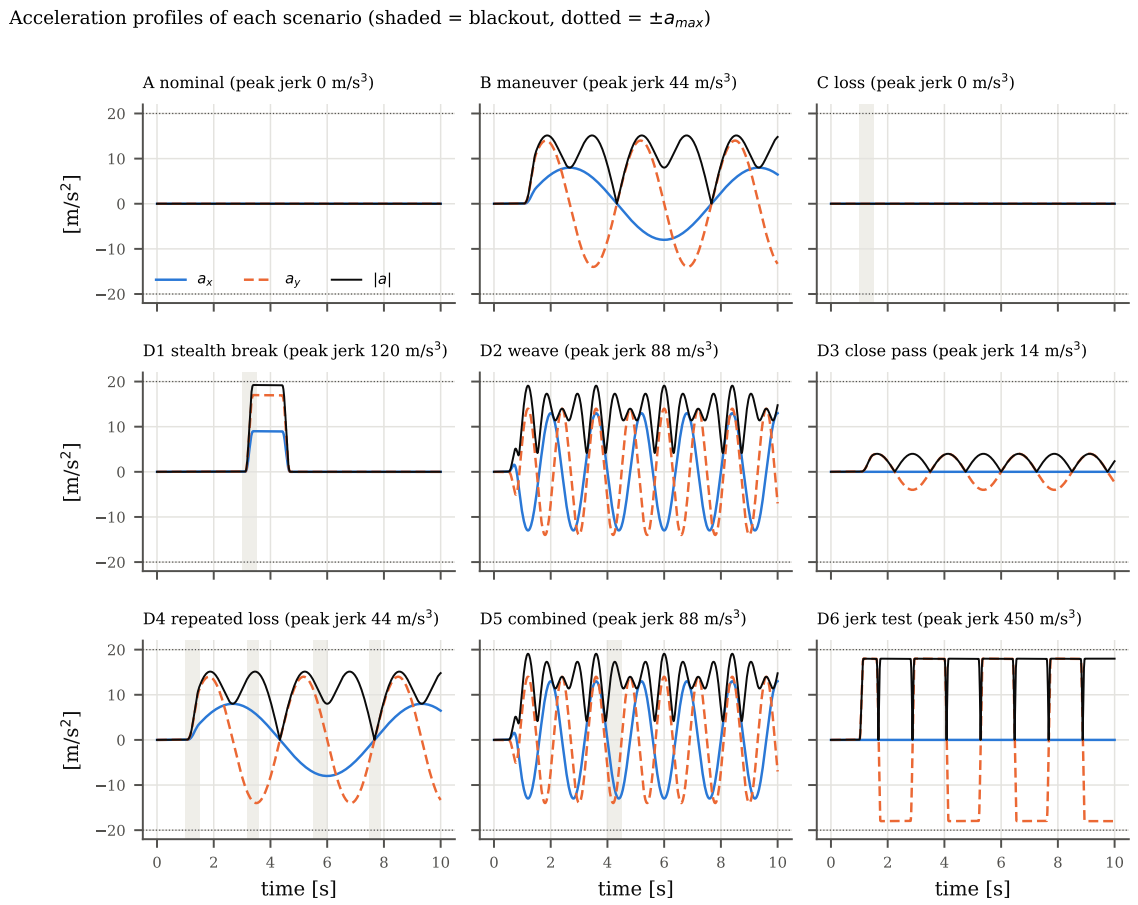


Figure 6: Acceleration profiles and peak jerk of each scenario. Shading marks blackouts.

4.3 Monte Carlo engagements

What we did. Each seed draws one random admissible engagement:

- initial range $U(1500, 2000)$ m, elevation $U(8°, 55°)$, speed $U(70, 150)$ m/s, and an inbound heading within $\pm 40°$;
- maneuver onset at $U(0.5, 3)$ s, then per axis a sum of three sinusoids (frequency $U(0.1, 1)$ Hz, random phase and amplitude), plus with probability 0.5 a smooth "break" pulse;
- the whole profile scaled to a peak of $U(0.4, 1) \times 0.97\, a_{\max}$;
- one blackout with onset $U(1, 8.5)$ s and duration $U(0.1, 0.5)$ s, and with probability 0.3 a second one.

Inadmissible draws are rejected. Radar noise uses an independent stream per seed.

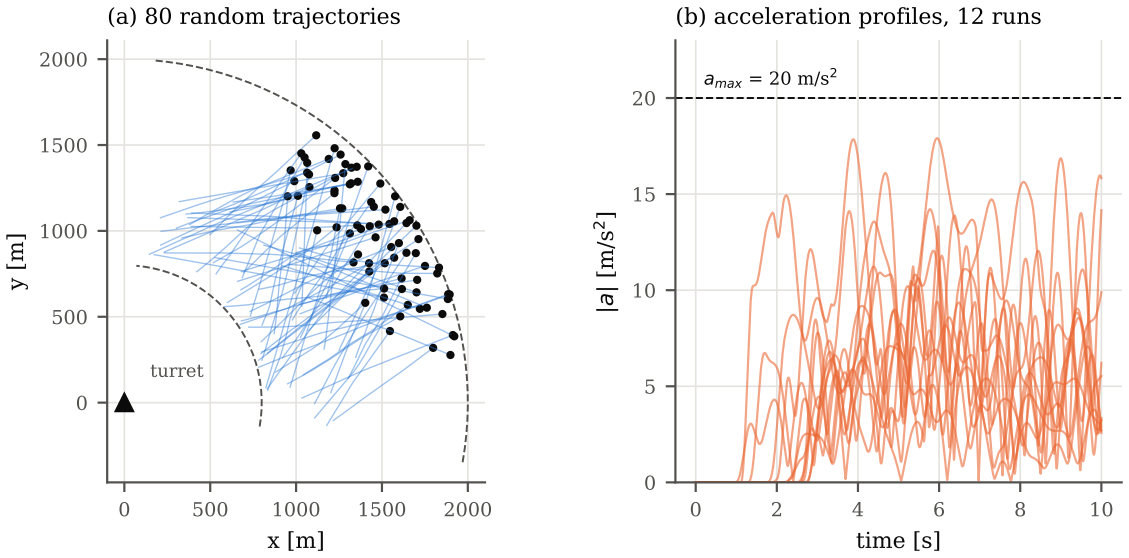


Figure 7: (a) Eighty sampled Monte Carlo trajectories. (b) Acceleration magnitude of twelve runs.

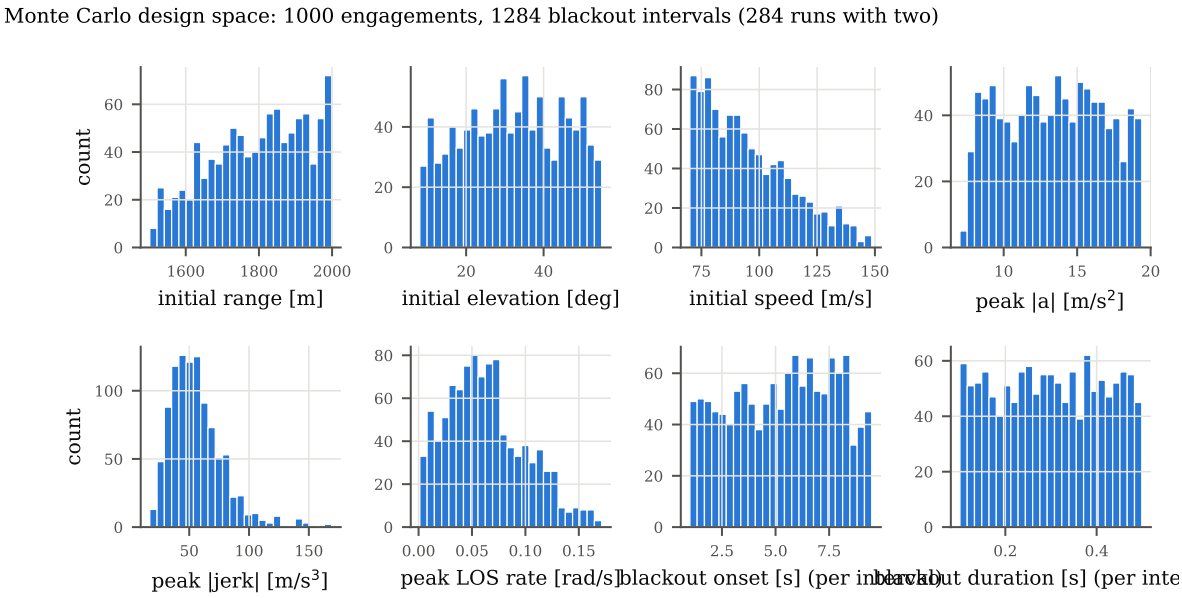


Figure 8: Realised design space over the 1 000 baseline engagements.

Observations.

- The 1 000 engagements span initial ranges of 1500–2000 m, elevations of 8–55°, speeds of 70–150 m/s, peak acceleration of 7.0–19.4 m/s^2, peak jerk of 17–169 m/s^3 and peak LOS rate up to 0.171 rad/s.
- There are 1 284 blackout intervals (716 runs with one, 284 with two), with onsets spread over 1–9.5 s and durations over 0.1–0.5 s (Fig. 8).
- The generator's jerk tops out near 170 m/s^3, well below D6's 450 m/s^3. This is why the stress noise ensembles (Section 7.2) are needed alongside the Monte Carlo.

4.4 Statistical methodology

- *Proportions* (interception, pulse hits): Wilson score 95% intervals. Pulse hit rates also get a cluster bootstrap over engagements (10 000 resamples), because pulses within one engagement are correlated.
- *Continuous metrics*: percentile bootstrap 95% intervals.
- *Design comparisons*: **paired** on identical seeds (same trajectory, blackouts and noise), using
  - Wilcoxon signed-rank tests (Pratt zero handling) for first-hit time and pointing RMS;
  - a sign-flip permutation test (20 000 permutations) for per-run missed pulses;
  - an exact McNemar test for interception;
  - Holm–Bonferroni correction within each metric family.
- *Estimator consistency*: the normalised error $z = e/\hat{\sigma}$ is compared with $\mathcal{N}(0, 1)$.
- *Firing calibration*: a reliability diagram and Brier score for P_{hit}.
- All random draws use fixed seeds.

5 Results I: Individual Scenarios

What we did. We ran each scenario once at 1 kHz with the frozen design. Each figure shows four panels:

1. true, estimated and t_p-predicted elevation and the turret angle;
2. cross-range pointing error, with the ± 2 m band and each pulse marked;
3. motor voltage;
4. the predicted hit probability P_{hit}.

Table 3: Scenario results (single realisation, 1 kHz controller).

ID	Description	First hit [s]	Pulses hit	Pointing RMS [m]	Time in R_{eng}
A	nominal	1.00	9/9	0.32	100%
B	maneuvering	1.00	9/9	0.53	100%
C	radar loss 1.0–1.5 s	0.98	10/10	0.41	100%
D1	stealth break	0.71	10/10	0.40	100%
D2	weave	1.02	9/9	0.99	96.3%
D3	close pass to 83°	1.27	9/9	0.32	100%
D4	four blackouts	0.97	9/9	0.59	100%
D5	weave + blackout	1.16	9/9	0.69	100%
D6	jerk test	1.10	8/9	0.78	98.1%

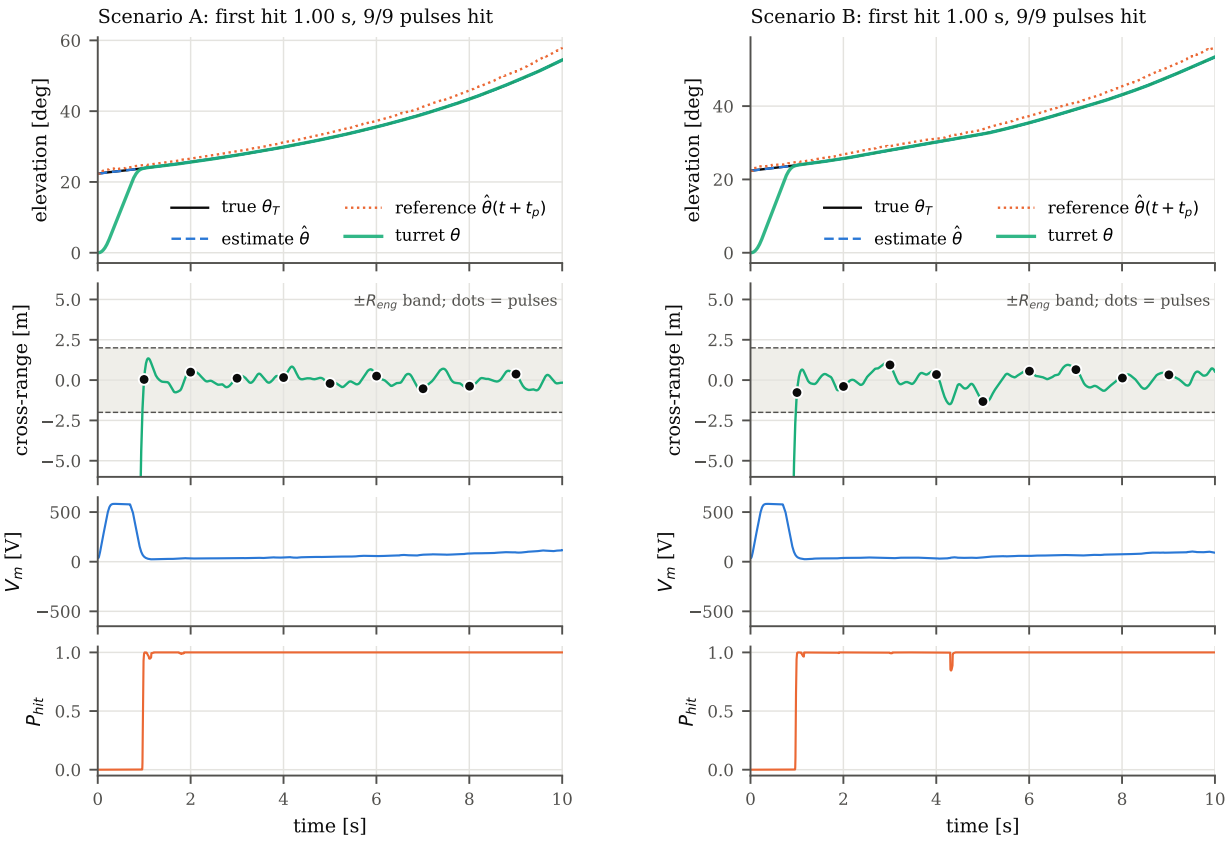


Figure 9: Scenario A (left) and Scenario B (right).

Observations.

- **Acquisition (A, B).** The turret starts at 0° with the target at 22°. The drive saturates at about 583 V for about 0.7 s while the reference generator's rate limit (0.58 rad/s) governs the slew. It then settles onto the target without overshoot and fires the first pulse at 1.00 s.
- **The t_p lead.** The orange predicted reference runs visibly *ahead* of the true elevation by about t_p throughout, while the turret (green) sits on the true elevation. This is the cancellation of H1 made visible.
- **A, steady state.** Pointing error stays within ±0.8 m. Voltage rises slowly from about 30 to 110 V as the closing target's LOS rate grows. Every pulse lands within 0.6 m of the centre.
- **B, maneuvering.** The sinusoidal maneuver causes excursions up to 1.5 m, and P_{hit} dips to 0.85 at 4.3 s, coinciding with an acceleration reversal. No pulse is fired during the dip. All 9 pulses hit.

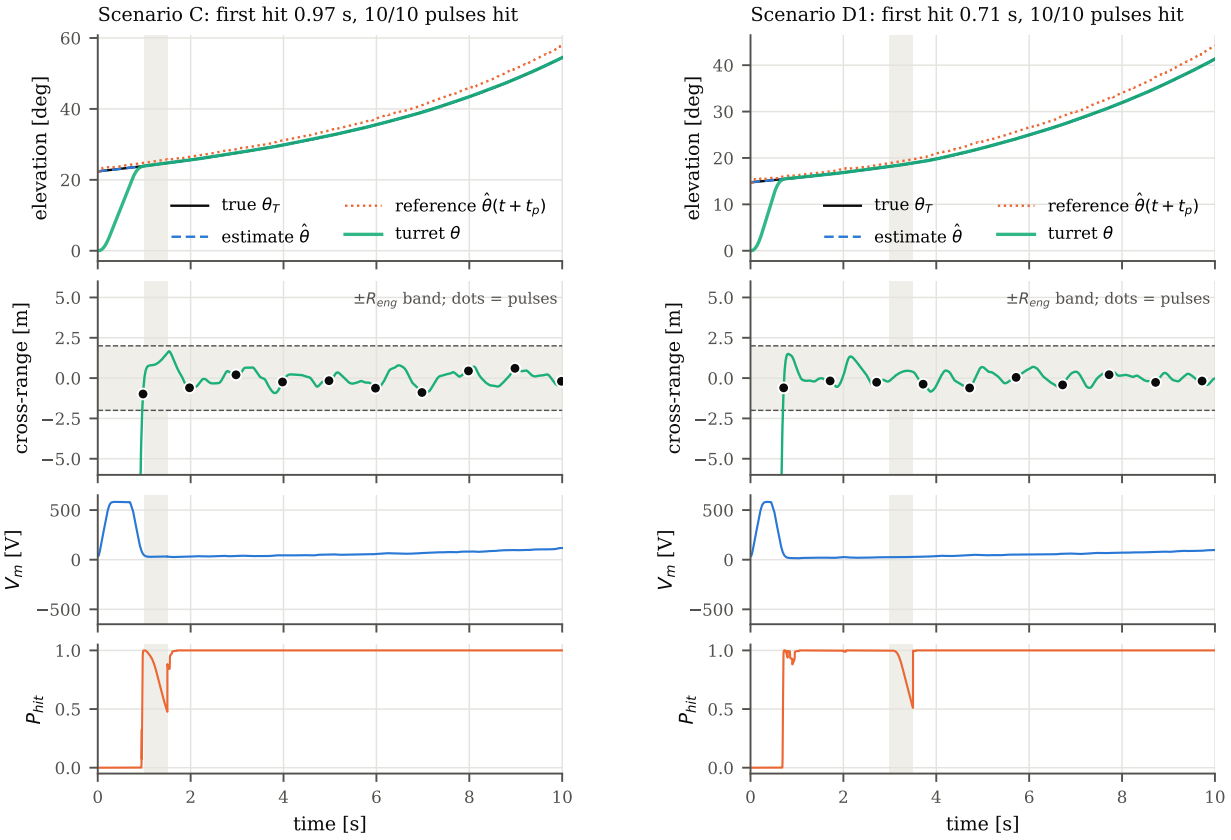


Figure 10: Scenario C (left) and stress scenario D1, stealth break (right). Shading marks the blackout.

Observations.

- **C.** The blackout starts at 1.0 s, just as the turret finishes acquisition. The first pulse fires at 0.98 s, *before* the loss. During the loss P_{hit} decays smoothly to about 0.35, reflecting the coasting covariance, and the firing rule holds fire. Pointing error briefly reaches 2.6 m at 1.5 s, because the track was only 1 s old when coasting began. After recovery the error returns inside 1 m within about 0.4 s, and every later pulse hits.
- **D1.** The target pulls up at about 19 m/s^2 *while unobserved*. The coasted estimate keeps the actual pointing error within about ± 0.5 m because the maneuver ramps in over 0.3 s, yet P_{hit} honestly falls to about 0.5 by the end of the loss. Fire was held: the pulse cadence (2.72 s, then 3.72 s) skips the blackout. All 10 pulses hit. The first hit comes at 0.71 s because the initial slew is only 15°.

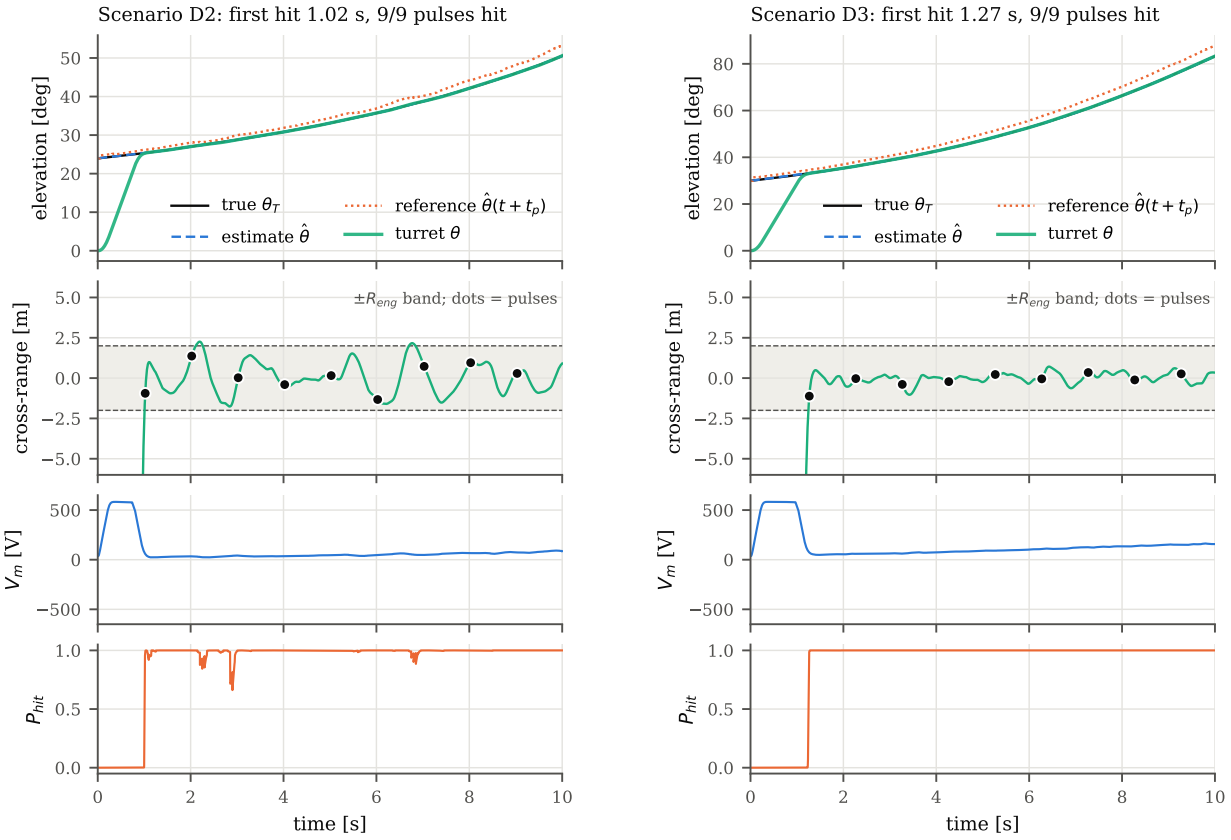


Figure 11: Stress scenarios D2, weave (left) and D3, close pass (right).

Observations.

- **D2.** This is the hardest pointing case. The near-$a_{\max}$ weave drives the error to ± 2.2 m *between* pulses, at about 2.2, 2.7 and 6.8 s, where P_{hit} drops to 0.67–0.85. The fire-now-or-wait rule never fires at those moments: every pulse lands within 1.4 m. This is the clearest demonstration of the firing logic (H4).
- **D3.** The elevation climbs to 83° and the LOS rate peaks near the end, so the voltage rises to about 160 V. This is the highest steady demand of any scenario, still far below the 600 V limit. Tracking is the tightest of all (about ± 0.6 m), and P_{hit} stays near 1.

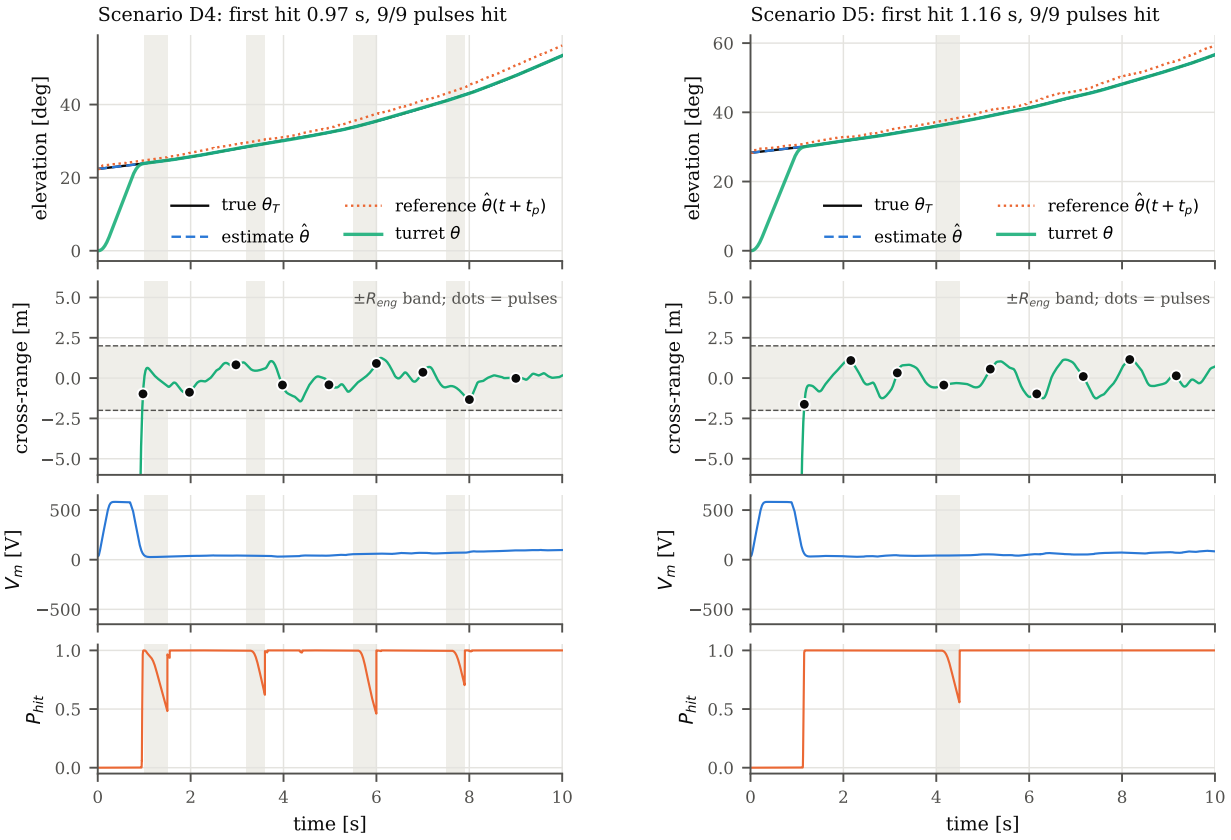


Figure 12: Stress scenarios D4, four blackouts (left) and D5, weave plus blackout (right).

Observations.

- **D4.** Each of the four blackouts produces the same signature: P_{hit} decays to 0.47–0.70, fire is held, and P_{hit} snaps back to about 1 at the first returning measurement. Pointing error stays within ± 1.4 m throughout, and all 9 pulses hit.
- **D5.** A pulse fires at about 4.15 s, *inside* the blackout. The rule allowed it because P_{hit} was still above 0.95 early in the loss, and it hit at 0.4 m. This is the "fire just after the loss starts" behaviour working as intended (H5).

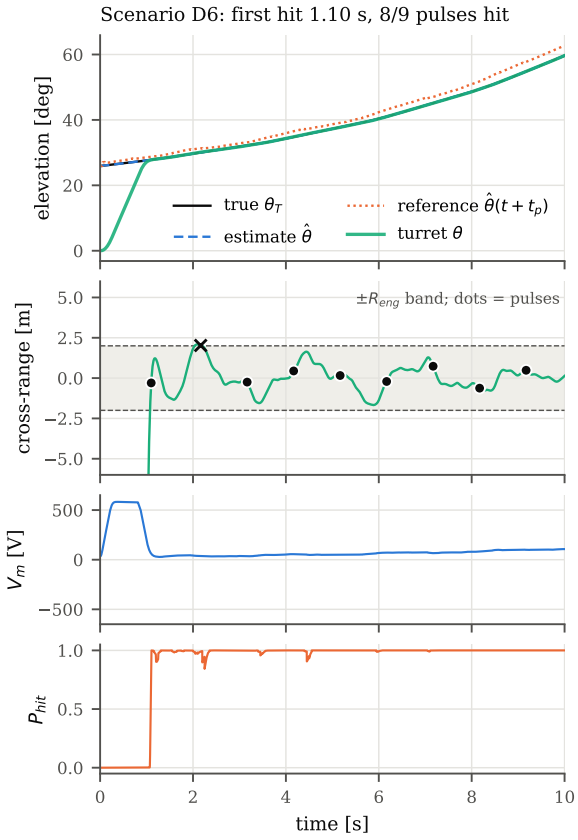


Figure 13: Stress scenario D6, jerk test.

Observations.

- D6 reverses $\pm 18\,\mathrm{m/s^2}$ every 1.2 s. The order-4 estimator follows each reversal, holding 98.1% of the time within R_{eng}.
- In this one realisation, the second pulse, at 2.1 s just after the first reversal, was fired at $P_{\mathrm{hit}} = 0.9901$ and missed by 2 cm: the estimate error (+0.93 m, about 2.4σ) and the turret lag (+1.09 m) added up in the same direction.
- Across 50 independent noise realisations of D6 (Section 7.2), the final design missed *none* of its D6 pulses. This miss is a single unlucky draw, not a systematic weakness.

6 Results II: Monte Carlo Campaign (1 000 engagements)

6.1 Interception and engagement accuracy

Table 4: Final design over 1 000 random admissible engagements (100 Hz).

Metric	Value	95% interval
Interceptions	1000/1000	[99.62%, 100%]; failure $< 0.38\%$
Pulse hit rate	99.96% (4 misses / 9 197)	[99.91, 99.99]% (cluster bootstrap)
First-hit time, mean	1.214 s	[1.190, 1.239] s
First-hit time, p95 / max	1.83 s / 2.14 s	
Pointing RMS, median / p95	0.559 m / 0.828 m	
Time within R_{eng}	99.40%	
Peak voltage	589 V	limit 600 V

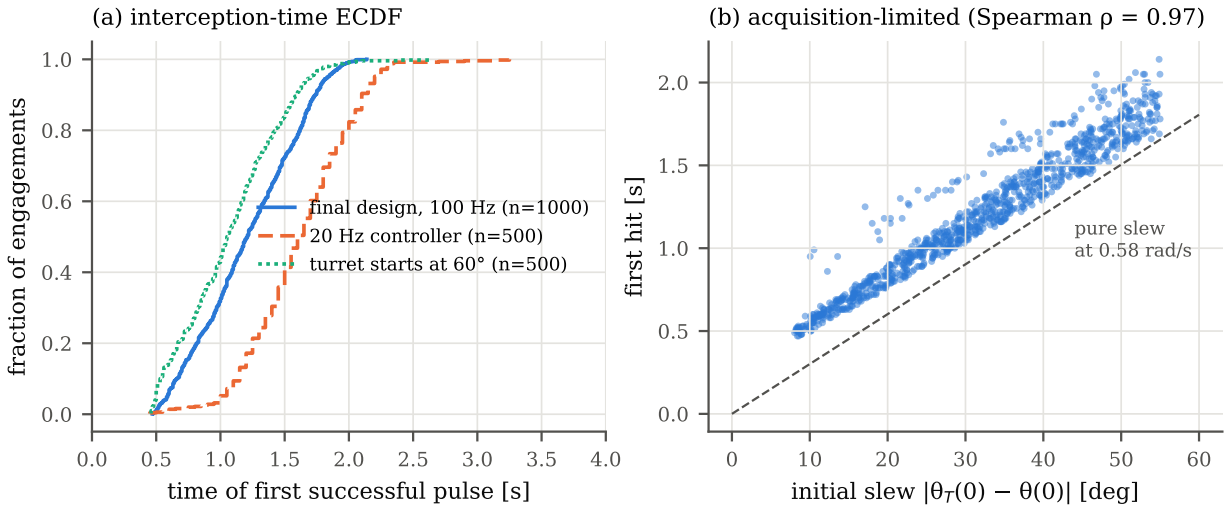


Figure 14: (a) Interception-time ECDFs for the final design (100 Hz), a 20 Hz controller and a turret starting at 60°. (b) First-hit time against the initial slew angle; the dashed line is a pure slew at the 0.58 rad/s limit.

Observations.

- Every engagement was intercepted. 50% are intercepted by about 1.2 s and all by 2.14 s (Fig. 14a).
- Interception time is almost entirely explained by the initial slew angle (Fig. 14b, Spearman $\rho = 0.97$). A least-squares fit gives

$$t_{\text{hit}} \approx 0.263\,\text{s} + 0.0296\,\tfrac{\text{s}}{\text{deg}} \times \text{slew}, \qquad R^2 = 0.93,\ \text{residual s.d. } 0.10\,\text{s}.$$

  The slope corresponds to a 0.589 rad/s slew, matching the 0.58 rad/s reference-generator limit.
- The intercept (about 0.26 s) is settling plus the 8-update firing gate. Targets needing under 10° of slew are intercepted at 0.51 s on average.
- First-hit time is *not* significantly correlated with maneuver intensity (RMS acceleration $\rho = 0.05$, $p = 0.15$), peak jerk ($\rho = 0.03$) or blackout time ($\rho = 0.05$, $p = 0.09$).
- Starting the turret at 60° shortens interception (mean 1.07 s), because most targets sit between 8° and 55°. A 20 Hz controller lengthens it (mean 1.62 s).

What it shows. Interception time is set by turret kinematics, not by estimation or by the target. It can be *predicted* for an unseen case from its geometry (used in Section 10).

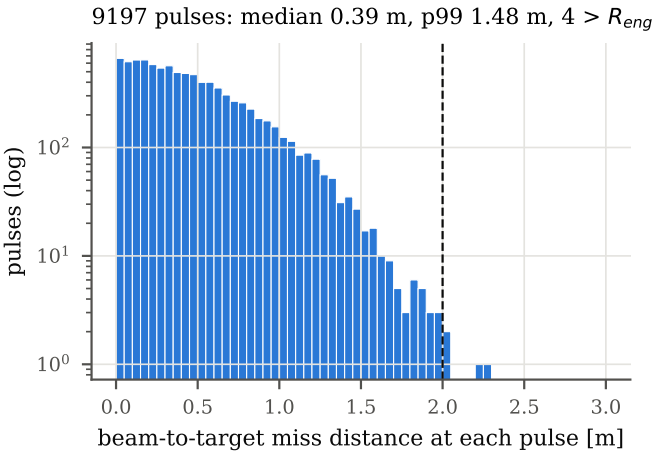


Figure 15: Beam-to-target miss distance at all 9 197 pulses (log scale).

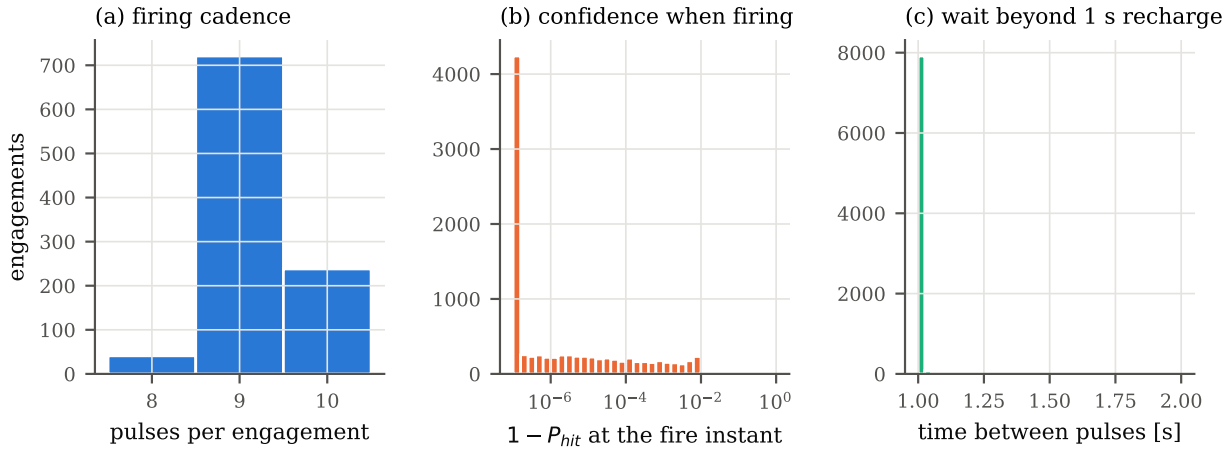


Figure 16: (a) Pulses per engagement. (b) $1 - P_{\text{hit}}$ at the fire instant. (c) Time between consecutive pulses.

Observations.

- **Miss distances** decay roughly exponentially (Fig. 15): median 0.39 m, 90th percentile 0.95 m, 99th percentile 1.48 m. Only 4 of 9 197 pulses exceed 2 m, all by 1–26 cm, and all are tracking shots.
- **Cadence.** Engagements fire 8–10 pulses (mean 9.2; Fig. 16a), close to the maximum allowed by the 1 s recharge after acquisition. 65% of follow-up pulses fire within 10 ms of the recharge completing, and 95% within 10 ms (Fig. 16c), so the wait-for-a-better-shot option is used sparingly.
- **Confidence.** 89.7% of pulses were fired at $P_{\text{hit}} \geq 0.999$ (Fig. 16b). 211 of 9 197 pulses were coasting shots (early in a blackout), and none of them missed.

What it shows. The firing rule converts tracking accuracy into hits with almost no waste, and it does so without slowing the cadence (supports H4).

6.2 Pointing accuracy over time

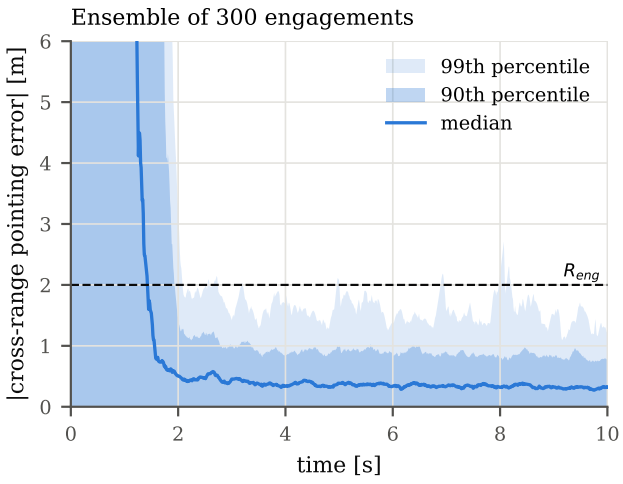


Figure 17: Cross-range pointing error over 300 logged engagements: median, 90th and 99th percentiles.

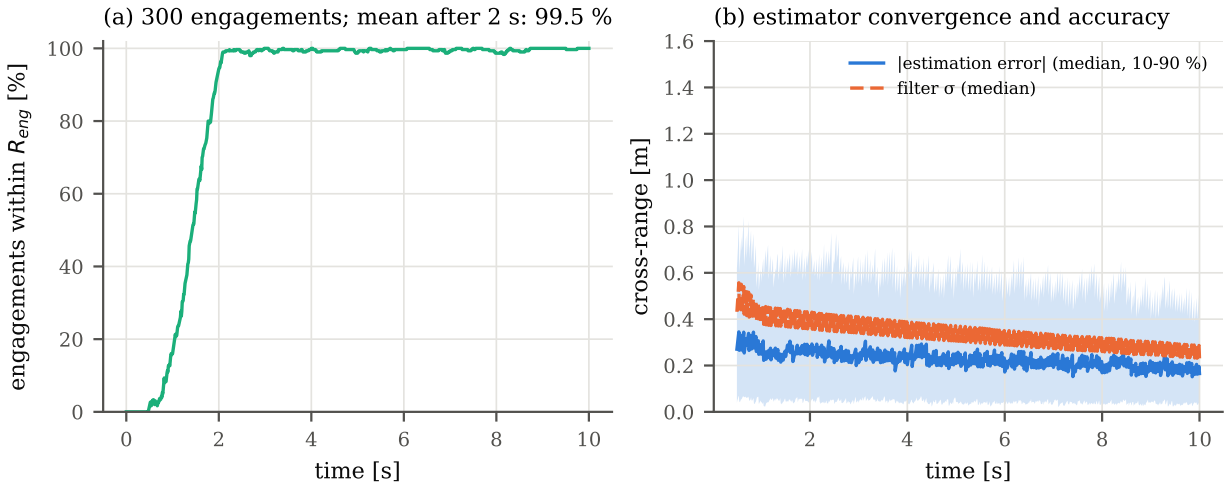


Figure 18: (a) Fraction of engagements pointing within R_{eng} over time. (b) Estimation error and filter σ over time (medians, 10–90% band).

Observations.

- The median pointing error drops below 2 m at 1.43 s, and the 90th percentile at 1.93 s (Fig. 17). After 2 s the median settles near 0.35 m and the 90th percentile below 1 m.
- The 99th percentile exceeds 2 m only in brief spikes, which correspond to blackout recoveries.
- After 2 s, 99.5% of engagement-time is within R_{eng} (Fig. 18a).
- The estimator converges within about 1 s. Afterwards the median cross-range estimation error is 0.22 m against a median filter σ of 0.32 m (Fig. 18b), far below the 0.7 m cross-range noise of a single radar sample at 2 km.

What it shows. Pointing error is set by the estimator, about 0.2–0.4 m, as H1 predicts: the t_p lead and the loop lag add essentially nothing in steady tracking.

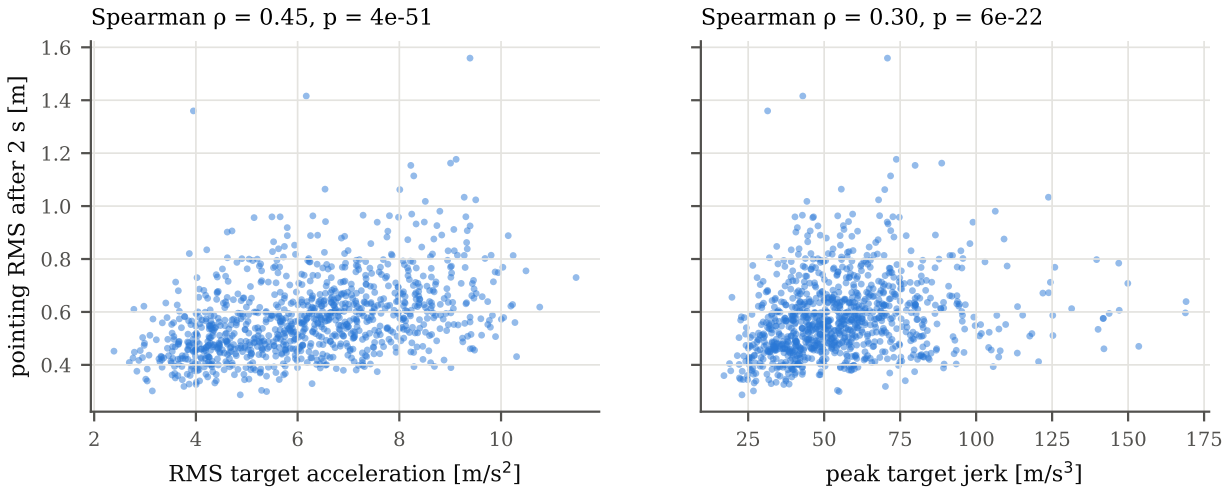


Figure 19: Pointing RMS against RMS target acceleration and peak target jerk.

Observations.

- Pointing accuracy degrades gracefully and monotonically with maneuver intensity: Spearman $\rho = 0.45$ with RMS acceleration and $\rho = 0.30$ with peak jerk, both highly significant (Fig. 19).

- Even the hardest engagements stay below 1.6 m RMS. The few outliers at low acceleration are engagements with long or double blackouts early in the track.

6.3 Behaviour across radar loss

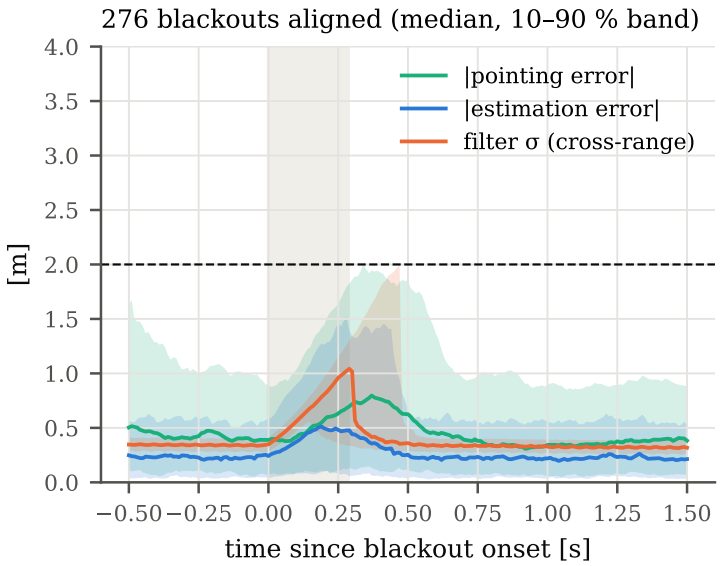


Figure 20: Errors and filter σ for 276 blackouts aligned at onset (median, 10–90% band; median blackout duration shaded).

Observations.

- **Before the loss**, the median pointing error is about 0.4 m and the median σ about 0.35 m.
- **During the loss**, the filter's σ grows fastest, to about 1.0 m at 0.3 s, because it allows for every admissible maneuver. The actual estimation error grows more slowly (median about 0.5 m), and pointing error lags it (median peak about 0.8 m at about 0.35 s).
- The 90th-percentile band reaches about 2 m only at the end of the longest losses. This is consistent with the $\frac{1}{2}a_{\max}t^2 = 2.5$ m worst-case drift in 0.5 s.
- **On recovery**, the first measurement collapses σ immediately, and pointing returns to its pre-loss level within about 0.4 s, set by the reference generator's triple pole at -20 rad/s.

What it shows. The system degrades gracefully, its uncertainty grows honestly (and conservatively), and it recovers fast (supports H5).

6.4 Estimator consistency and firing calibration

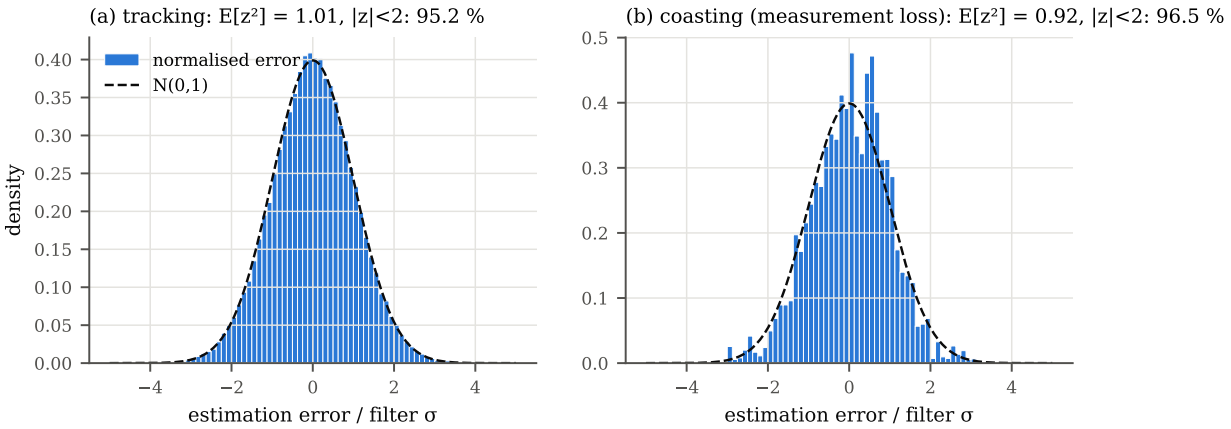


Figure 21: Normalised estimation error $z = e/\hat{\sigma}$ against $\mathcal{N}(0,1)$: (a) tracking, (b) coasting.

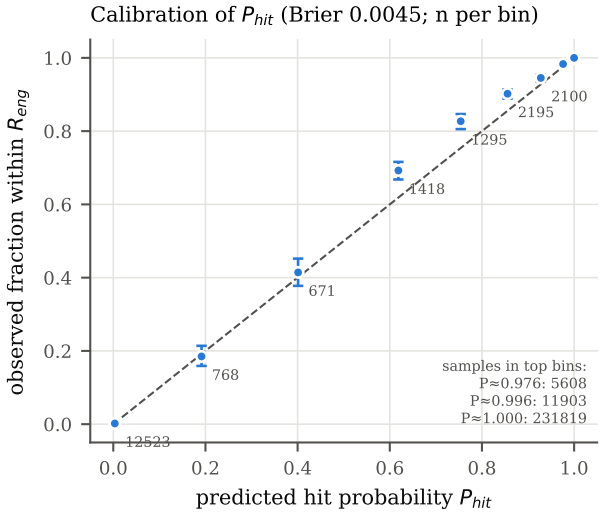


Figure 22: Reliability diagram of the predicted hit probability (about 270 000 controller samples; Wilson 95% intervals).

Observations.

- **Tracking.** The normalised error is almost exactly standard normal: $E[z^2] = 1.008$ (ideal 1), with 95.2% of samples within $|z| < 2$ (ideal 95.4%).
- **Coasting.** The distribution is narrower: $E[z^2] = 0.92$, 96.5% within $|z| < 2$. During losses the filter is slightly conservative, which is the safe direction.
- **Calibration.** Observed hit frequency follows the predicted P_{hit} in every bin, with a Brier score of 0.0045 (Fig. 22). In the mid-range bins the observed frequency is *higher* than predicted (0.62→0.69, 0.75→0.83).
- At the 9 197 fire instants, the mean predicted P_{hit} implied 8.6 expected misses; 4 occurred.

What it shows. The probabilities driving the firing decision are honest and slightly conservative (supports H2).

7 Results III: Design Decisions under Paired Statistics

7.1 One-factor ablations (500 paired seeds)

What we did. Each design choice was reverted or perturbed one at a time and run on the same 500 seeds as the final design.

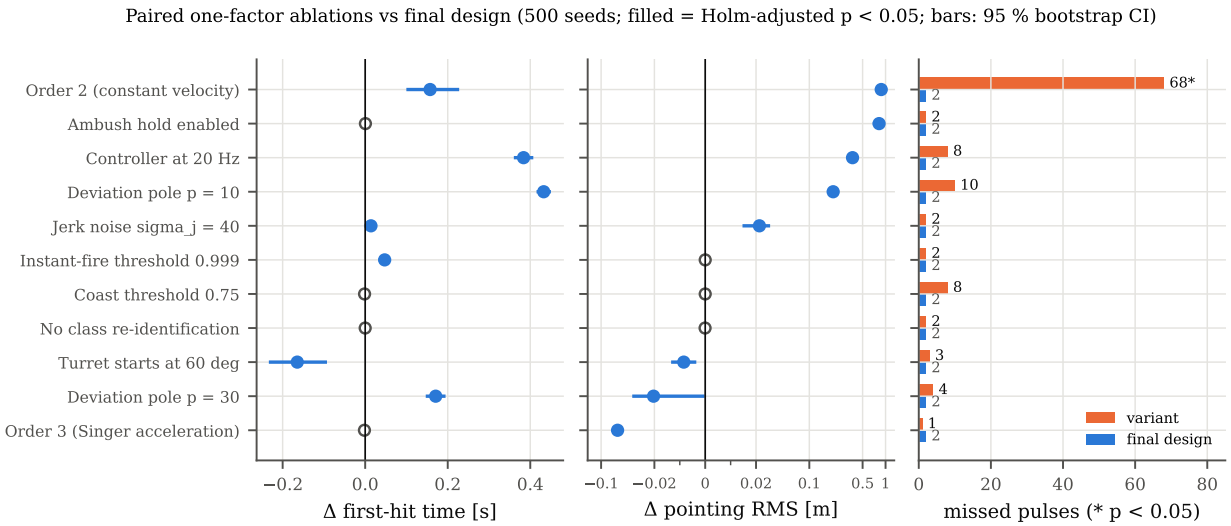


Figure 23: Paired ablations: mean paired difference (variant minus final) with 95% bootstrap CI; filled markers have Holm-adjusted $p < 0.05$. Right: missed pulses.

Table 5: Paired ablations (variant minus final; Holm-adjusted p).

Variant	Δ first hit [s], p	Δ pointing RMS [m], p	Misses final→var.	Verdict
Order 2 (constant velocity)	$+0.157$, 5×10^{-43}	$+0.878$, 10^{-74}	2→68; 3 int. lost	strongly worse
Deviation pole $p = 10$	$+0.433$, 2×10^{-82}	$+0.206$, 10^{-82}	2→10	strongly worse
Ambush hold on	$+0.001$, 1	$+0.822$, 6×10^{-51}	2→2	rejected
Controller at 20 Hz	$+0.384$, 6×10^{-78}	$+0.369$, 10^{-82}	2→8	slower, still 500/500
Deviation pole $p = 30$	$+0.171$, 10^{-31}	-0.021, 3×10^{-49}	2→4	slower for a marginal gain
Instant-fire 0.999	$+0.047$, 2×10^{-36}	0	2→2	slower, no gain: rejected
Jerk noise $\sigma_j = 40$	$+0.014$, 0.0013	$+0.022$, 3×10^{-10}	2→2	slightly worse
Coast threshold 0.75	-0.002, 0.63	0	2→8 ($p = 0.65$)	more misses (n.s.)
No class re-identification	0, 1	0, 0.77	2→2	no measurable effect
Order 3 (Singer)	-0.002, 1	-0.061, 5×10^{-70}	2→1	better on smooth MC
Turret starts at 60°	-0.165, 2×10^{-5}	-0.008, 0.0024	2→3	shorter slew

Observations.

- **Estimator order matters most.** A constant-velocity filter (order 2) more than doubles pointing RMS, multiplies misses by 34 and loses 3 interceptions.
- **The deviation pole is a genuine trade-off.** At $p = 10$, corrections are absorbed slowly, so interception is 0.43 s later and pointing 0.21 m worse. At $p = 30$, acquisition is 0.17 s slower, because the cascade saturates longer, for only 0.02 m better pointing. $p = 20$ sits at the knee.
- **Two of our own ideas did not survive.** The ambush hold (park the beam where every maneuver class must cross) worsened pointing by 0.82 m with no extra hits. Class re-identification at recovery had no measurable effect, because the Kalman update with the coasted covariance already performs that correction. We report both rather than claim them.

- **Firing thresholds.** 0.999 instead of 0.99 delays interception with identical misses. A coast threshold of 0.75 instead of 0.95 quadruples misses on these seeds, though not significantly at this sample size.

7.2 Stress-scenario noise ensembles

What we did. The Monte Carlo maneuvers are smooth, so we re-ran all nine scenarios A–D6 with 50 independent radar-noise realisations each: 450 paired engagements per variant, at 100 Hz and at 20 Hz.

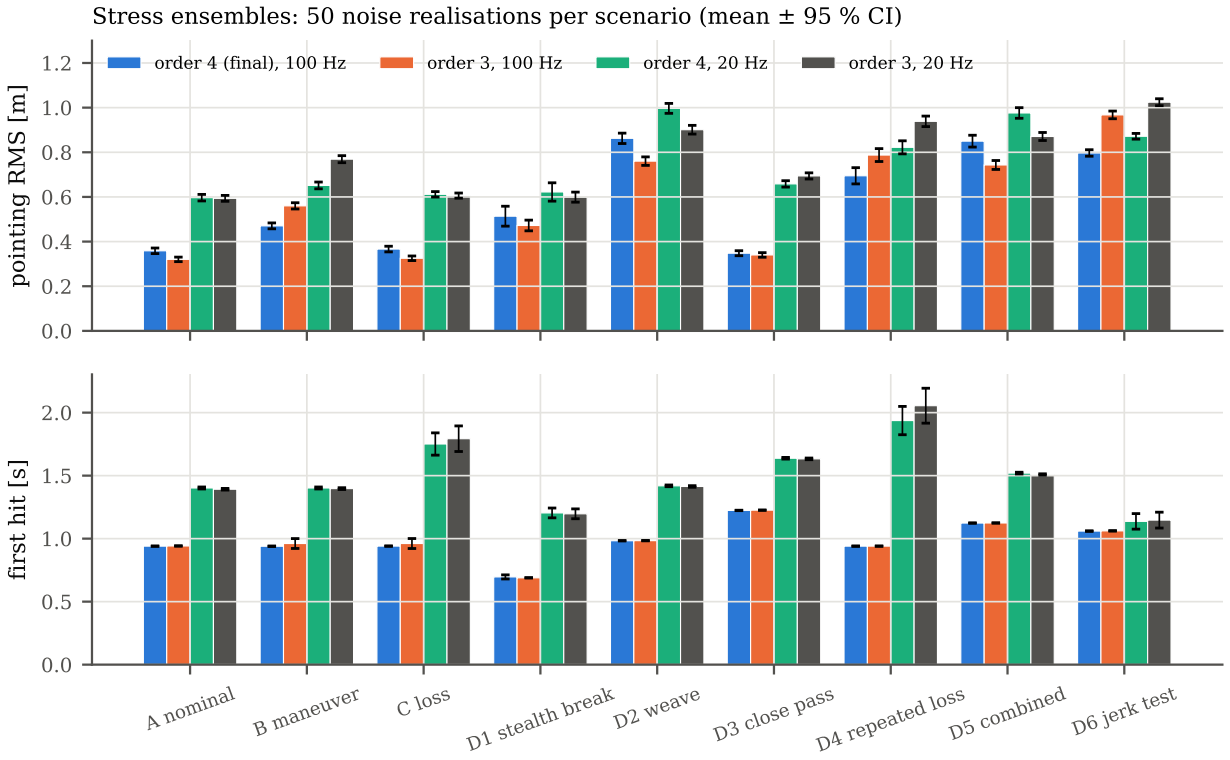


Figure 24: Per-scenario pointing RMS and first-hit time over 50 noise realisations (mean ± 95% CI): order 4 (final) against order 3, at 100 Hz and 20 Hz.

Table 6: Stress ensembles: 9 scenarios × 50 noise realisations.

Variant	Intercepted	Pulse hit rate [95% CI]	Misses	First hit [s]	Pointing RMS [m]
Final (order 4), 100 Hz	450/450	99.90% [99.75, 99.96]	4	0.983	0.585
Order 3, 100 Hz	450/450	99.88% [99.71, 99.95]	5	0.988	0.586
Instant-fire 0.999	450/450	99.90% [99.75, 99.96]	4	1.013	0.585
Coast threshold 0.75	450/450	99.90% [99.75, 99.96]	4	0.983	0.585
Final (order 4), 20 Hz	450/450	99.61% [99.35, 99.76]	15	1.490	0.756
Order 3, 20 Hz	450/450	99.27% [98.95, 99.50]	28	1.504	0.777

Observations.

- **Every engagement in every variant was intercepted.**
- **Per-scenario pattern** (Fig. 24): order 3 points better on smooth or weaving motion (A, C, D1, D2, D5; by 0.04–0.11 m). Order 4 is better wherever the acceleration *switches*: B by 0.09 m, D4 by 0.09 m, and the jerk test D6 by 0.17 m. All of these differences have $p < 0.005$.
- **At 100 Hz** the two orders tie on average (ΔRMS +0.002 m, $p = 0.82$).

- **At 20 Hz**, where fewer corrections per radar sample amplify model mismatch, order 4 is significantly better overall: order 3 adds +0.021 m RMS ($p = 1.1 \times 10^{-4}$) and nearly doubles the misses (28 vs 15).
- The stricter instant-fire threshold again only delays interception (+0.029 s, $p = 3 \times 10^{-33}$).
- First hit is acquisition-limited in every scenario, except D4 at 20 Hz, where the first blackout (1.0–1.5 s) overlaps acquisition and pushes the first hit to about 1.9 s.

What it shows. Order 4 is not uniformly better: on smooth targets order 3 is slightly more accurate. But order 4 wins exactly on the cases a hidden adversarial scenario is likely to contain (maneuver switches, high jerk, repeated loss) and at low controller rates. For an unseen evaluation, it is the robust choice (supports H3 in its robust form).

8 Results IV: Controller Rate, Timing and Implementation

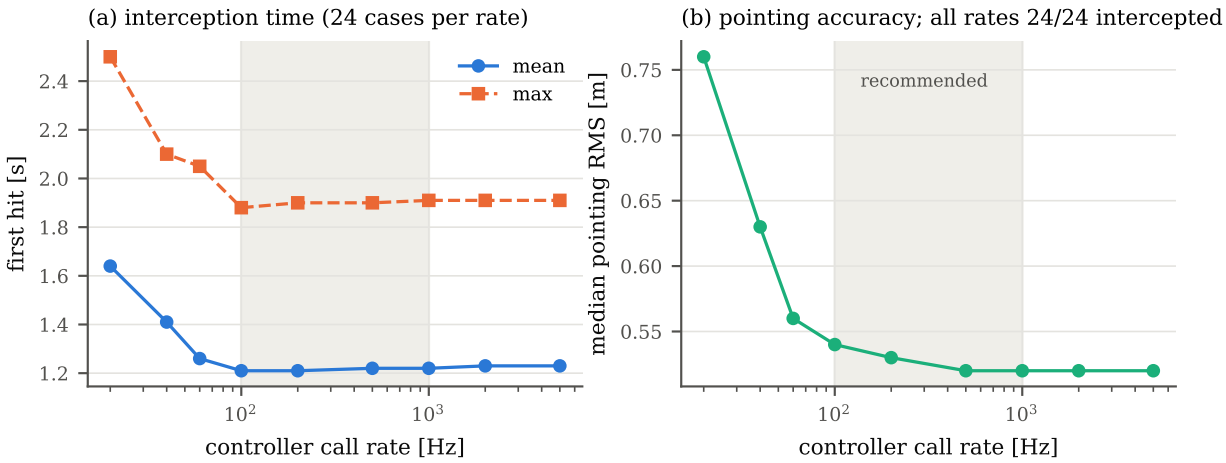


Figure 25: Interception time and pointing accuracy against controller call rate (24 cases per rate: A–D6 and 15 Monte Carlo seeds).

Table 7: Controller-rate study.

Rate [Hz]	Intercepted	Pulses hit	First hit mean / max [s]	Pointing RMS (median) [m]
20	24/24	201/201	1.64 / 2.50	0.76
40	24/24	215/215	1.41 / 2.10	0.63
60	24/24	215/215	1.26 / 2.05	0.56
100	24/24	219/219	1.21 / 1.88	0.54
200	24/24	220/220	1.21 / 1.90	0.53
500	24/24	220/221	1.22 / 1.90	0.52
1000	24/24	220/221	1.22 / 1.91	0.52
2000	24/24	220/221	1.23 / 1.91	0.52
5000	24/24	220/221	1.23 / 1.91	0.52

Observations.

- Every target is intercepted at every rate from 20 Hz to 5 kHz.
- Performance improves up to about 100–200 Hz and is flat above. The single miss at ≥ 500 Hz is the D6 reversal shot of Section 5.

- With forward-Euler plant integration at 1 ms (controller 100 Hz), Scenario B still intercepts at 0.95 s. Forward Euler is unstable above 3.34 ms because of the 8.9 Hz mode, so the README requires a plant step of at most 1 ms.
- The MATLAB `turret_controller.m` costs 0.51 ms per call (median 0.33 ms), so real-time use is practical up to about 1 kHz.
- Replaying 12 logged Python runs through the MATLAB controller reproduces the voltage within 3×10^{-9} V with identical fire commands, so the submitted code behaves exactly as the experiments report.

9 Results V: Robustness to the Evaluation Interface

What we did. The instructor evaluates with his own script and hidden cases, which we never see. Besides unseen trajectories, the script may also deliver measurements and integrate the plant differently from our harness. We therefore ran the frozen design under eight plausible conventions, 300 engagements each on identical seeds:

- **I1** (reference): NaN with `meas_valid`=0 between radar ticks, 100 Hz.
- **I2**: the last measurement *held*, with `meas_valid`=0 between ticks and during blackouts.
- **I3**: the last measurement *repeated with `meas_valid`=1* on every call, including during blackouts (stale duplicates).
- **I4, I5**: irregular radar timing, where a sample is taken at the call time once 50 ms have elapsed, with call rates of 30 Hz and 333 Hz (steps that do not divide 50 ms).
- **I6**: the turret starting at a random angle $U(-10°, 85°)$.
- **I7**: forward-Euler plant integration at 1 ms.
- **I8**: the reference convention at 1 kHz.

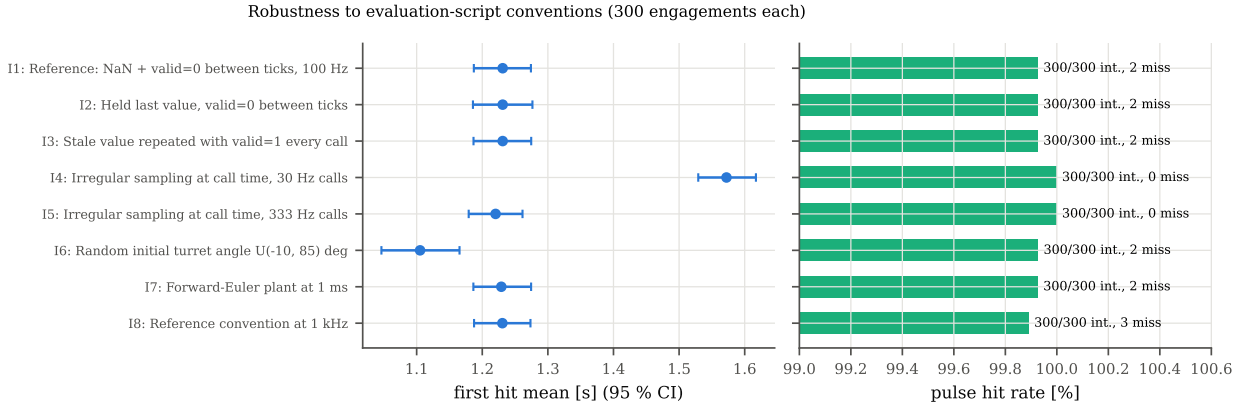


Figure 26: Frozen design under eight evaluation-script conventions (300 engagements each, identical seeds): mean first-hit time with 95% CI, and pulse hit rate.

Table 8: Robustness to evaluation-script conventions. Δ is paired against I1 (Wilcoxon p).

ID	Convention	Intercepted	Pulse hit rate (misses)	First hit mean / max [s]	Δ first hit [s], p
I1	reference, 100 Hz	300/300	99.93% (2)	1.231 / 2.05	–
I2	held value, `valid`=0	300/300	99.93% (2)	1.231 / 2.05	0, 1
I3	stale value repeated, `valid`=1	300/300	99.93% (2)	1.231 / 2.05	0, 1
I4	irregular sampling, 30 Hz	300/300	100.00% (0)	1.572 / 2.30	$+0.341$, 7×10^{-45}
I5	irregular sampling, 333 Hz	300/300	100.00% (0)	1.220 / 1.97	-0.011, 10^{-25}
I6	random initial angle $-10°..85°$	300/300	99.93% (2)	1.105 / 2.80	-0.126, 3×10^{-4}
I7	forward-Euler plant, 1 ms	300/300	99.93% (2)	1.229 / 2.05	-0.002, 0.99
I8	reference, 1 kHz	300/300	99.89% (3)	1.231 / 1.96	0, –

Observations.

- **All 2 400 engagements were intercepted** under every convention.
- **Held or stale measurements are harmless.** I2 and I3 are bit-for-bit identical to I1, so the duplicate-measurement guard and the reliance on `meas_valid` work. A script that keeps re-asserting `meas_valid`=1 with an old sample, even through a blackout, cannot corrupt the track.
- **Irregular radar timing is handled.** With samples taken whenever 50 ms have elapsed, at call steps that do not divide 50 ms (I4, I5), there were no misses at all. At 30 Hz interception is 0.34 s later, an effect of the coarse call rate (as in Section 8), not of the irregular timing. At 333 Hz it matches the reference.
- **Any initial turret angle works.** With the turret starting anywhere in $[-10°, 85°]$ (I6), the worst first hit was 2.80 s, consistent with the slew model of Section 10. The higher pointing RMS of this variant comes from 11 runs that started at 76°–85°, whose long downward slews were still under way past the 2 s mark where RMS starts counting.
- **Plant integration and rate.** Forward-Euler at 1 ms (I7) and a 1 kHz call rate (I8) change nothing material.

What it shows. The design does not depend on our harness's conventions, so the unseen evaluation script should reproduce the performance reported here (supports H6).

10 Final Evaluation: Expected Performance on Unseen Cases

The instructor's evaluation cases are hidden, and the design was frozen before any of them could be seen. This section therefore states, from our own evidence, what the frozen design should be expected to do on them, and where the expectation stops.

10.1 What the hidden cases share with our tests

The specification guarantees that every evaluation case satisfies:

- the operating envelope: $800 \le r \le 2000\,\text{m}$, $|\mathbf{v}| \le 150\,\text{m/s}$, $|\mathbf{a}| \le 20\,\text{m/s}^2$, with smooth bounded maneuvers in x and y;
- measurement-loss bounds: onset at or after 1 s and duration at most 0.5 s.

Our tests cover that envelope from three directions:

- random sampling across it (1 000 + 500 + 500 engagements, Fig. 8);

- deliberately adversarial corners (D1–D6, each with 50 noise realisations): maneuvering inside a blackout, near-$a_{\max}$ weaving, a close pass to 83°, four blackouts, and 450 m/s^3 jerk;
- eight evaluation-script conventions (Section 9).

No parameter was tuned per scenario. One parameter set was used for all of them.

10.2 Quantitative expectations

1. **Interception.** Across all test sets the frozen design faced **2 900** distinct engagements at its nominal settings (Monte Carlo at 100 Hz, at 20 Hz and from a 60° start, plus stress ensembles at 100 and 20 Hz), and intercepted all of them. The 95% upper bound on the probability of failing to intercept an admissible engagement is **0.13%** (Wilson; the rule of three gives 0.10%). The interface experiment adds 2 400 more engagements under other conventions (Section 9).
2. **Interception time.** This is predictable from geometry (Fig. 14b):

$$t_{\text{hit}} \approx 0.26 + 0.0296 \times \left|\theta_T(0) - \theta_{\text{turret}}(0)\right|_{\text{deg}} \text{ s} \quad (\pm 0.20\,\text{s},\ 95\%).$$

   For a turret starting at 0° and targets between 8° and 55°, expect 0.5–1.9 s (mean about 1.2 s). The worst conceivable slew, 95° from −10° to +85°, gives about 3.1 s. A blackout overlapping acquisition delays the first hit until after recovery, by up to 0.5 s.
3. **Engagement accuracy.** Expect a pulse hit rate of 99.8–99.99% at 100 Hz–1 kHz, and about 99.6–99.8% at 20 Hz. Typical pulses land 0.4 m from the target (99th percentile 1.5 m).
4. **Consistency across cases.** Pointing RMS varies smoothly with maneuver intensity (0.3–0.6 m on gentle targets, 0.8–1.0 m on near-$a_{\max}$ weaves). Its 95th percentile over all Monte Carlo engagements is 0.83 m. No case type produced a qualitatively different failure mode.

10.3 Interface requirements the evaluation script must meet

- Plant integration with a step $\leq$ 1 ms, or `ode45`. With larger forward-Euler steps the *plant itself* becomes numerically unstable.
- Controller called at $\geq$ 20 Hz; 100 Hz–1 kHz is recommended. Any step works, because the controller reads Δt from t.
- Nothing else is needed. The controller resets itself when t decreases or $t = 0$, ignores non-new or NaN measurements, and works from any initial turret angle.

10.4 Where the evidence stops

The expectations above hold for admissible engagements as specified. They are not evidence for:

- maneuvers sharper than tested (jerk above about 450 m/s^3, i.e. near-instant acceleration switches);
- blackouts longer than 0.5 s, after which drift exceeds R_{eng} for any controller;
- targets above the +85° elevation stop, which are mechanically unreachable;
- a plant model that differs from the specified one, where robustness rests on the analytical margins (PM 68°, GM 16 dB, $M_s = 1.21$) rather than on tests.

11 Verdicts and Conclusions

Table 9: Verdict on each hypothesis of Section 1.

	Evidence	Verdict
H1	Turret sits on the true elevation while the reference leads by t_p (Section 5). Pointing error (0.2–0.4 m) matches estimation error (0.22 m median). $\theta - y_d$ is under 3 cm. Exact cancellation exists only to order 4 and is fixed by deviation feedback (Fig. 4).	**Supported**
H2	$E[z^2] = 1.008$ tracking, 0.92 coasting. Calibrated P_{hit} (Brier 0.0045). 4 misses observed vs 8.6 predicted.	**Supported**
H3	Order 2: 3 interceptions lost and 34× the misses. Order 4 vs 3: better on maneuver switches (B, D4, D6) and at 20 Hz; slightly worse on smooth targets.	**Supported in robust form**
H4	99.96% pulse accuracy. Cadence at the recharge limit (median gap 1.01 s). D2 shows fire withheld at bad moments. 0.999 threshold only slower.	**Supported**
H5	Graceful degradation (median 0.8 m peak), honest σ growth, 0.4 s recovery. No late blind shots. Coasting shots 211/211 hit.	**Supported**
H6	2 900/2 900 interceptions (failure $< 0.13\%$); geometry-predictable timing; robust to eight interface conventions.	**Supported within the envelope**

Conclusion. The evidence supports the thesis.

- Treating the prescribed prediction horizon as a lag to be cancelled, rather than a lead to be applied, yields a design in which the turret tracks the target's *present* elevation. The time-shift property makes the cancellation exact, and the stability of the Taylor sections bounds it at order four.
- Pointing error is then limited only by estimation, about 0.2–0.4 m against a 2 m engagement radius.
- A jerk-state estimator matched to the plant's relative degree, an exact plant-inverse feedforward and a calibrated firing rule turn that accuracy into hits: 99.96% of pulses, with interception time set by turret kinematics alone.
- The same frozen design intercepted every one of 2 900 admissible engagements, including adversarial maneuvers inside radar blackouts, and is insensitive to how an evaluation script delivers measurements.

Two of our own ideas, the ambush hold and class re-identification, were tested and discarded or demoted by the same evidence. The order-4 choice is justified by robustness on maneuver switches, not by uniform superiority. The design's limits are those of physics, not of the controller: turret slew rate bounds interception time, and the reachable set of a maneuvering target bounds what any controller can guarantee late in a long blackout.

A Reproducing This Report

```
cd python
bash scripts/run_all_experiments.sh       # all core experiments
python scripts/interface_robustness.py --n 300
python scripts/plot_report_extras.py
# MATLAB (matlab/): parity_check, time_controller, run_scenarios
```

Raw results are written to `python/results/`: the Monte Carlo statistics, the stress and interface reports, and the rate-study table.